

《多元函数积分学》 检测题参考答案

一、选择题 (15 分)

1. C 2. C 3. A 4. B 5. D

二、填空题 (15 分)

1. $-\int_1^2 dx \int_{1-x}^0 f(x, y) dy$ 2. $4\pi t^2 f(t^2)$ 3. $12a$ 4. $\sqrt{2}\pi$ 5. 0

三、(8分) 解 原式 $= \int_0^{\frac{\pi}{2}} d\theta \int_0^{\sqrt[4]{2}} r^3 \sin \theta \cos \theta [1+r^2] dr$ (4分) $= \int_0^{\frac{\pi}{2}} \sin \theta \cos \theta d\theta \int_0^{\sqrt[4]{2}} r^3 [1+r^2] dr$ (6分)
 $= \frac{1}{2} (\int_0^1 r^3 dr + \int_1^{\sqrt[4]{2}} 2r^3 dr) = \frac{3}{8}$ (8分)

四、(8分) 解 1. $D_{xy}: x^2 + y^2 \leq 2 \Rightarrow D_{xy}: 0 \leq r \leq \sqrt{2}; 0 \leq \theta \leq 2\pi$.

$V = \iint_{D_{xy}} \left(\sqrt{3-r^2} - \frac{1}{2}r^2 \right) r dr d\theta$ (2分) $= \int_0^{2\pi} d\theta \int_0^{\sqrt{2}} \left(\sqrt{3-r^2} - \frac{1}{2}r^2 \right) r dr$ (3分) $= \frac{\pi}{3} (6\sqrt{3} - 5)$ (4分)

2. $S = S_1 + S_2 = \iint_{S_1} dS + \iint_{S_2} dS = \iint_{D_{xy}} \frac{\sqrt{3}}{\sqrt{3-r^2}} r dr d\theta + \iint_{D_{xy}} \sqrt{1+r^2} r dr d\theta$ (6分)
 $= \int_0^{2\pi} d\theta \int_0^{\sqrt{2}} \frac{\sqrt{3}}{\sqrt{3-r^2}} r dr + \int_0^{2\pi} d\theta \int_0^{\sqrt{2}} \sqrt{1+r^2} r dr$ (7分) $= \frac{16}{3} \pi$ (8分)

五、(8分) 解 作辅助曲线 $\overline{BA}$, 则 $\int_L = \oint_{L+\overline{BA}} - \int_{\overline{BA}}$,

$\oint_{L+\overline{BA}} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \iint_D (y^2 + x^2) dx dy$ (3分) $= \int_0^\pi d\theta \int_0^a r^2 \cdot r dr = \frac{\pi a^4}{4}$, (5分)

$\overline{BA}: y=0, dy=0, \int_{-a}^a \sin x dx = 0$, (7分) $\therefore \int_L = \oint_{L+\overline{BA}} - \int_{\overline{BA}} = \frac{\pi a^4}{4}$. (8分)

六、(8分) 解 $M = \iiint_V (x^2 + y^2 + z^2) dx dy dz$ (1分) $= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} \sin \varphi d\varphi \int_0^{2R \cos \varphi} \rho^4 d\rho$ (2分) $= \frac{32}{15} \pi R^5$, (3分)

$M_z = \iiint_V z(x^2 + y^2 + z^2) dx dy dz$ (4分) $= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} \sin \varphi \cos \varphi d\varphi \int_0^{2R \cos \varphi} \rho^5 d\rho$ (5分) $= \frac{8}{3} \pi R^6$, (6分)

$\bar{z} = \frac{M_z}{M} = \frac{5}{4} R$, (7分) $\therefore$ 重心为 $(0, 0, \frac{5}{4} R)$ (8分).

七、(8分) 解 $\frac{\partial P}{\partial y} = \frac{-2xy}{(x^2 + y^2)^2}$, $\frac{\partial Q}{\partial x} = \frac{2axy}{(x^2 + y^2)^2}$, 当 $a = -1$ 时, $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, (2分)

(1) 当 L 不包围原点时, $x^2 + y^2 \neq 0$, $\oint_L \frac{xdx + ydy}{x^2 + y^2} = 0$. (4分)

(2) 当 L 包围原点时, 作 $l: x = \delta \cos \theta, y = \delta \sin \theta$,

$\oint_L = \oint_l = \int_0^{2\pi} \frac{\delta \cos \theta (-\delta \sin \theta) + \delta \sin \theta (\delta \cos \theta)}{\delta^2} d\theta = 0$, (7分) $\therefore$ 当 $a = -1$ 时, 有 $\oint_L \frac{xdx + ydy}{x^2 + y^2} = 0$. (8分)

八、(8分) 解 $f(tx, ty) = t^{-2} f(x, y)$ 两边对 t 求导, 令 $t=1$ 得: $xf'_x(x, y) + yf'_y(x, y) = -2f(x, y)$, (2分)

$$\frac{\partial Q}{\partial x} = -f(x, y) - xf'_x(x, y), \frac{\partial P}{\partial y} = f(x, y) + yf'_y(x, y), (4分)$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = -f(x, y) - xf'_x(x, y) - [f(x, y) + yf'_y(x, y)] = 0. (6分) \Rightarrow \oint_L = 0. (8分)$$

九、(10分) 解 $\overrightarrow{OM} = \{a, b, c\}$, $\overrightarrow{OM} : \frac{x}{a} = \frac{y}{b} = \frac{z}{c} = t$, (1分) 且 $x=0$ 时, $t=0$, $x=a$ 时, $t=1$, (2分)

$$W = \int_{\overrightarrow{OM}} \vec{F} \cdot d\vec{l} = \int_{\overrightarrow{OM}} yzdx + xzdy + xydz (4分) = \int_0^1 (btcta + ctatb + atbt c) dt = 3abc \int_0^1 t dt = abc, (5分)$$

三角形所在平面为: $\frac{x}{1} + \frac{y}{2} + \frac{z}{3} = 1$,

令 $G = abc + \lambda(\frac{a}{1} + \frac{b}{2} + \frac{c}{3} - 1)$, (7分) 由 $G_a = 0$, $G_b = 0$, $G_c = 0$, (9分) $\Rightarrow a = \frac{1}{3}$, $b = \frac{2}{3}$, $c = 1$. (10分)

十、(10分) 解 作辅助平面 $S_1: z=0$, $(x^2 + y^2 \leq R^2)$, (1分)

$$\oiint_{S^- + S_1} \frac{axdydz + (z+a)^2 dxdy}{\sqrt{x^2 + y^2 + z^2}} = \frac{1}{a} \oiint_{S^- + S_1} axdydz + (z+a)^2 dxdy (3分) = \frac{1}{a} \iiint_V (3a + 2z) dV (5分) = \frac{3}{2} \pi a^3, (6分)$$

$$\frac{1}{a} \iint_{S_1} axdydz + (z+a)^2 dxdy = \frac{1}{a} \iint_{x^2 + y^2 \leq a^2} a^2 dxdy = \pi a^3, (8分)$$

$$I = \frac{1}{a} \iint_S = -\frac{1}{a} \iint_{S^-} = -\left(\frac{3}{2} \pi a^3 - \pi a^3\right) = -\frac{1}{2} \pi a^3. (10分)$$